P510/2

Physics

Paper 2

July – August 2026

3 Hours

UGANDA MUSLIM TEACHERS' ASSOCIATION

UMTA JOINT MOCK EXAMINATIONS – 2026

UGANDA ADVANCED CERTIFICATE OF EDUCATION

Physics

Paper 2

Practical

3 Hours

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of* **two** *compulsory items.*
- *Use* **blue** *or* **black** *ink when responding to tasks.*
- *Any work done in pencil except graphs and diagrams will not be scored.*
- *Scores are awarded mainly for a clear record of the observations actually made, for their suitability, accuracy and for the use made of them.*
- *Details on the question paper should not be repeated in the responses.*
- *Graph paper is provided.*
- *Silent non – programmable scientific calculators may be used.*

Item 1.

A small Technology startup company in Eastern Uganda is developing low – cost optical devices for rural schools. One of their products is a simple projection system that uses a convex lens to display enlarged images from printed materials onto a wall. Before mass production, the engineers need to assess the suitability of the Convex lens they have sourced by determining its power. The engineers also suggest that for suitability of the Convex lens to be used in the projection system, its power should be in the range **9.0 diopters to 12.0 diopters.**

Since it is just a startup company, it has no instruments which can be used to measure directly the power of the sourced Convex lens. The company manager has been directed to your school physics laboratory for help, but unfortunately, the laboratory technician is not feeling well, therefore he is unable to help.

However, he suggests that the power of the sourced Convex lens can be investigated by using two independent methods to ensure reliability and consistency.

One method involves the use of a plane mirror and a white screen with illuminated cross wires.

The second method involves the movement of the lens in order to observe a clear magnified image and a clear diminished image of an illuminated object (cross wires) on the screen for different distances between the illuminated object (cross wires) and the screen and then obtaining the displacement through which the lens is being moved, then through graphical analysis of the results.

The laboratory technician has also recommended that you as a physics student at advanced level, you can be of help.

The manager of the company has now approached you with the sourced Convex lens for help.

Task:

Carry out a scientific investigation to address the engineers' need.

Item 2.

A rapidly growing municipality has completed the construction of a modern community library intended to serve several surrounding schools. The library is expected to accommodate thousands of textbooks, reference books, and journals. Since most of the shelves will remain heavily loaded throughout the year, the local engineering firm responsible for furnishing the library has been asked to carryout scientific investigations about the stiffness of wood intended to be used in the construction of the wooden shelves to ensure that the shelves do not bend excessively after installation.

In order for the shelves not to bend excessively, the Young's modulus of wood to be used must be in the range 8.0×10^9 to 12×10^9 Pa.

The company has obtained timber from a local sawmill. Before the timber is approved for mass production, its stiffness (Young's Modulus). Because the actual shelves have not yet been manufactured, strips cut from the same timber have been prepared in the form of wooden metre rules, each having thickness, **t** and breadth **b**, and investigations about the value of Young's modulus are to be carried out scientifically.

You are provided with a sample of the wooden metre rule and the other apparatus required for the investigation.

Task:

Follow the instructions provided and use the result obtained to investigate whether the shelves will not bend excessively when the timber obtained by the company is used.

(a) Measure the breadth, **b** and thickness, **t** of the metre rule provided.

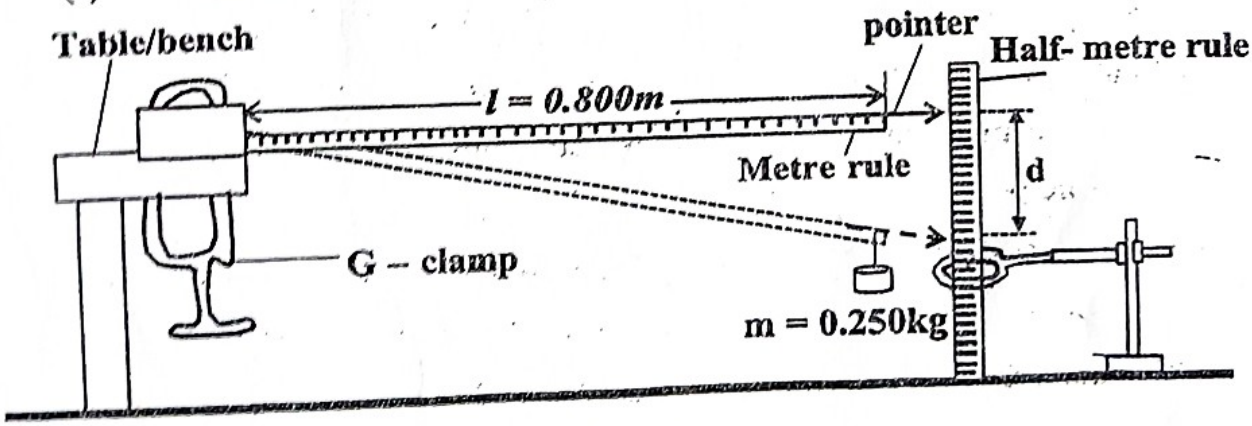


Figure 1

(b) Arrange the apparatus as shown in **Figure 1** above.

(c) Read and record the initial position, $\mathbf{P_o}$ of the pointer on the half metre rule.

(d) Suspend a mass $m = 0.250kg$ from the free end of the metre rule.

(e) Read and record the new position, $\mathbf{P_1}$ of the pointer on the half metre rule.

(f) Obtain the depression, d (in metres) of the metre rule.

(g) Calculate the value of $\mathbf{Y_1}$ using the expression: $Y = \frac{4gl^3m}{bt^3d}$

Part II

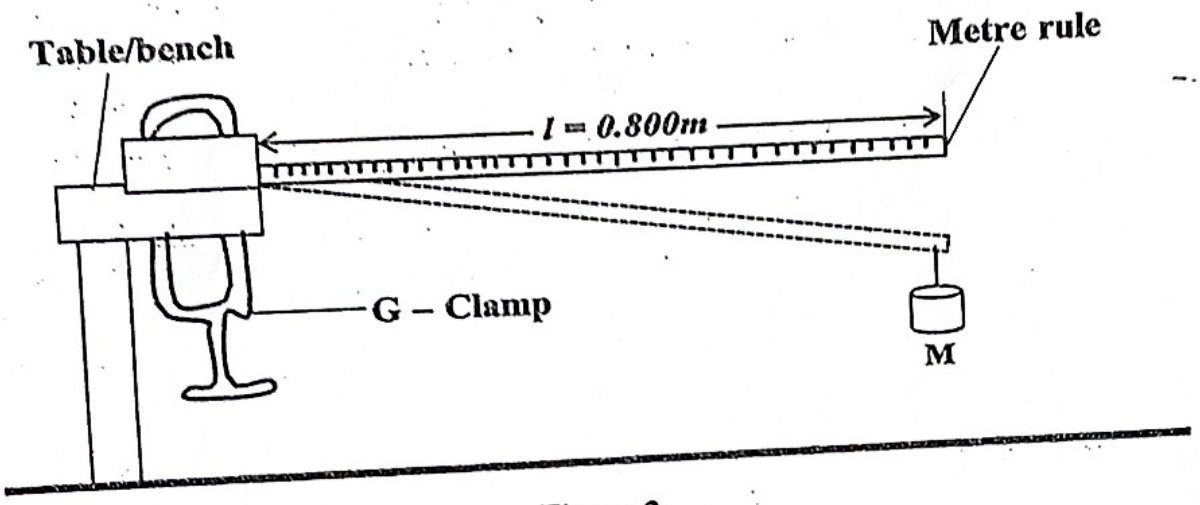


Figure 2

(a) Arrange the apparatus as shown in **figure 2** above.

(b) Suspend a mass **M = 0.050kg** from the free end of the metre rule.

(c) Depress the mass **M** through a small vertical distance and release it to oscillate.

(d) Measure and record the time for **20** oscillations.

(e) Find the period, **T** for one oscillation

(f) Repeat steps **(b)** to **(e)** for values of **M = 0.100, 0.150, 0.200, 0.250** and **0.300kg**.

(g) Tabulate the results including values of $\mathbf{T^2}$.

(h) State sources of errors and their corresponding precautions taken to minimize/mitigate them.

(i) Plot a graph of T^2 against M.

(j) Obtain the slope, S of the graph.

(k) Calculate the value of Y_2 from the expression $\frac{Y_2=16n^2t^3}{Sbt^3}$

(l) Determine the Young's modulus, Y of the wooden metre rule by calculating the average value Y_1 and Y_2.

(m) Provide a feedback in form of a recommendation or advise to the company engineer about the ability of the shelves to support the textbooks and other suggested material without bending excessively.

END